ENTERPRISE AIOPS PLATFORM • ARCHITECTURE SPECIFICATION

Enterprise AIOps Platform - Architecture Overview

Target Platform: Google Cloud Platform (GCP)

System of Record: ServiceNow ITSM

Version: 2.0 (Production Blueprint)

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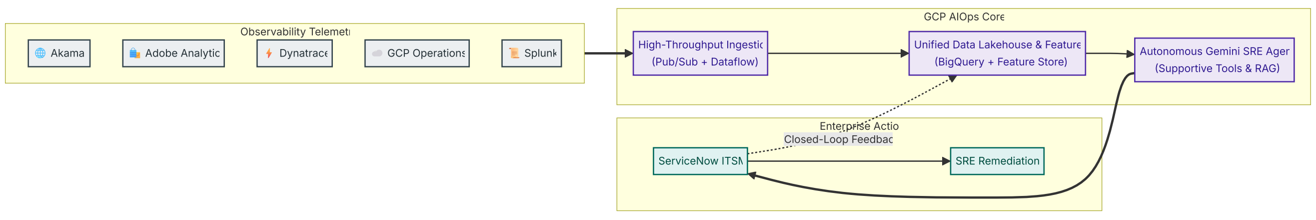
Enterprise AIOps Platform - High-Level Architecture Overview

1. Executive Summary & Business Objective

Modern global enterprise operations face extreme telemetry fragmentation. For a large multi-channel retail organization, operations span thousands of edge locations, extensive cloud footprints, and millions of customer transactions per hour.

The **Enterprise AIOps Platform** consolidates disparate observability streams from the SRE team's 5 core tools—**Google Cloud Platform (GCP)**, **Dynatrace**, **Akamai**, **Splunk**, and **Adobe Analytics**—into a centralized intelligence engine natively hosted on Google Cloud Platform. The platform correlates business transactions with technical performance to automate incident triage, accelerate root cause identification, and trigger proactive remediation in **ServiceNow**.



2. Core Architectural Pillars

2.1 Multi-Source Ingestion to GCP Core

While telemetry originates from diverse specialized tools across edge networks, cloud platforms, and enterprise data centers, **all centralized stream processing, analytical storage, and AI reasoning workloads run natively on Google Cloud Platform (GCP)**. This prevents vendor lock-in across tool providers while consolidating data for cross-domain ML models.

2.2 ServiceNow as the Definitive ITSM System of Record

ServiceNow is the enterprise system of record for all incident lifecycles, configuration item (CMDB) topologies, on-call schedules, and automated remediation audit logs. The AIOps layer communicates bidirectionally via standard ServiceNow REST APIs to enrich existing records, generate consolidated incidents, and bypass manual L1 triage.

2.3 Event-Driven Streaming & Elastic Scale

The platform is built on an event-driven architecture using **Cloud Pub/Sub** and **Cloud Dataflow (Apache Beam)**. It comfortably sustains normal loads of 150,000+ Events Per Second (EPS) and auto-scales to absorb peak promotional traffic (e.g., Black Friday / Cyber Monday surges exceeding 500,000 EPS) without message loss or latency degradation.

2.4 Actionable AI & Autonomous Incident Diagnostics

Rather than producing passive dashboards, the AI architecture utilizes the **Autonomous Gemini SRE Agent** with **Supportive Function Calling Tools** to execute tangible operational workflows:

- De-duplicating cascading alerts into unified incident signatures.
- Retrieving matching Standard Operating Procedures (SOPs) from **Vertex AI Vector Search**.
- Automatically executing non-destructive diagnostic queries and posting synthesized findings to ServiceNow tickets.
- Capturing closed-loop SRE feedback upon ticket closure to continuously evaluate and tune prompts.

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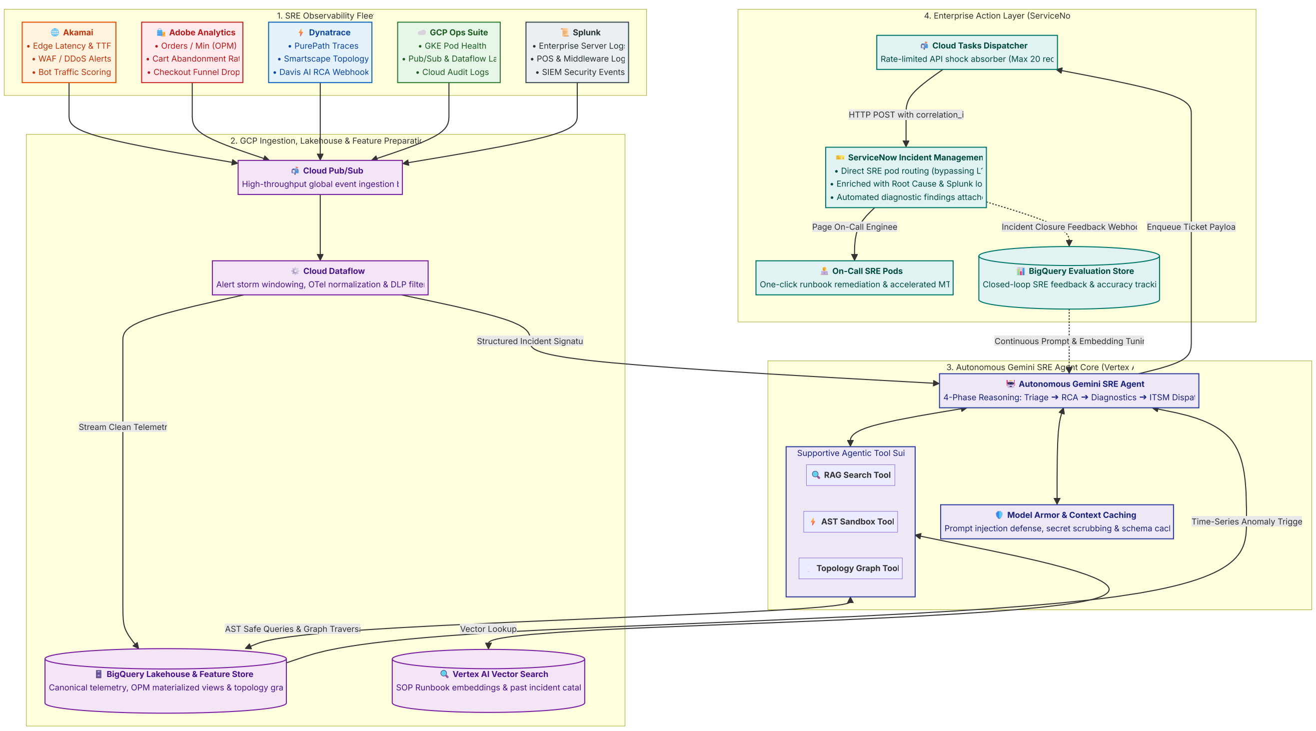
3. End-to-End Architectural Flow



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4. Architectural Domain Map & Detailed References

Architecture Domain	Scope & Responsibilities	Deep-Dive Specification
01. Ingestion Architecture	Transport protocols, Pub/Sub topologies, Dataflow Beam pipelines, Cloud DLP scrubbing, dead-letter queues, and unified telemetry matrix for all 5 observability tools.	• 01_ingestion/README.md • source_telemetry_matrix.md
02. Storage, Lakehouse & Data Processing	BigQuery dataset designs, table partitioning by ingestion time, Parquet on GCS cold storage tiering, stateful alert clustering, time-series resampling, and topology graph ETL.	• lakehouse_architecture.md • data_processing_and_feature_store.md
03. Intelligence & Reasoning	Autonomous Gemini SRE Agent runtime, 4-phase reasoning lifecycle, supportive function calling tools (RAG, AST Sandbox, Topology Graph), Model Armor guardrails, and context caching.	aiops_intelligence_layer.md
04. ITSM & Remediation	ServiceNow Incident Management integration, bidirectional lifecycle synchronization, CMDB topological sync, and automated diagnostic execution.	servicenow_integration.md